Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Mon, Sep 28, 2026 at 7:49 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

I have verified revision 3. The principal objection is resolved: the new proof of Proposition 7.5 replaces the Riccati step entirely.
The correction to the random-limit statement is ok, and your three replies to Sara are well directed.

Let me point you three narrow amendments remaining in my opinion. Two are consistency items, one is a scope item on pre-existing material.

1. The positivity of the rates is used but not stated.
The proof reads "Since the rates k_Π, k_ν are positive and $k_\bullet\delta t < 1$, $u>0$ and $E \succeq 0$."
But Proposition 7.5 invokes the setting of Corollary 3.6, whose hypotheses are $k_\Pi\delta t < 1$ and $k_\nu\delta t < 1$ and nothing more.
Those do not imply positivity, and both the Euler chain and the OU surrogate are well defined for a negative rate (explosive, but Gaussian with positive density on a finite horizon) so nothing in the setting supplies it.

Three points may help you choose the repair, because the sentence bundles two conclusions of different status and the
exact condition is available.

(a) $u > 0$ does follow from the stated hypothesis. Since $a_\bullet = 1 - k_\bullet\delta t$, the condition $k_\bullet\delta t < 1$
gives $a_\bullet > 0$ for either sign of $k_\bullet$: with $k_\Pi = -3$, $k_\nu = -2$, $\sigma_\nu = 1.3$, $\lambda = 0.5$, $M = 100$
one has $a_\Pi = 1.030$, $a_\nu = 1.020$, $u = 1.3318 > 0$, while $\kappa = -3.5917 < 0$.
Only $E \succeq 0$ needs the extra hypothesis.

(b) For $E \succeq 0$ the exact condition is $\kappa \geq 0$, and it is necessary as well as sufficient.
Sufficiency is immediate, since $\kappa_2 = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2 \geq 0$ always, being a sum of squares, so E is a sum of two PSD matrices. Necessity has a one-line witness: L is strictly lower triangular,
so $(Le_M)_m = \sum_{j<m}\delta_{jM} = 0$ for every $m \leq M$, i.e. $Le_M = 0$, whence

$$e_M^\top E\, e_M = \delta t\,\kappa\,(\mathbf{1}^\top e_M)^2 + \delta t^2\kappa_2\|Le_M\|^2 = \delta t\,\kappa.$$

So if $\kappa < 0$ the last increment alone certifies that E is not PSD, for every M.
There is no window of negative κ rescued by the $\delta t^2\kappa_2 L^\top L$ term.

(c) $\kappa \geq 0$ is strictly weaker than positivity of both rates and suffices for the whole proposition.
With $k_\Pi = 2$, $k_\nu = -0.5$, $\lambda = 0.5$, $\sigma_\Pi = \sigma_\nu = 1$ one has $\kappa = 1.75 > 0$ with a negative rate; over $(k_\Pi,k_\nu) \in \{(2,-0.5),(3,-1),(1,-0.2)\}$ and $M = 100, 400, 1600$.
I verified with no violations that $C_M \preceq (\delta t/u)I$, that $\mathbb{E}[V_M]$ lies between the two bounds of the proposition,
and that $\operatorname{Var}(V_M) \leq 2T\delta t/u^2$.

Either repair is one line. I would suggest stating $\kappa \geq 0$ as the hypothesis, with $k_\Pi, k_\nu > 0$ named as the case of interest;
the witness e_M shows the two conditions are separated by nothing, so stating the exact one costs nothing and is more informative.
Adding $k_\Pi, k_\nu > 0$ to Corollary 3.6 would also be entirely acceptable.

2. Table 3 retains the finite equality without the remainder.
The abstract, the introduction and (C1) now carry $1/\sigma_{\mathrm{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$ correctly.
The corresponding row of Table 3, on page 31, still prints the equality with no remainder and no refinement-limit label.
Please add the $O(\delta t)$ or label the equality as the refinement limit.

3. The χ^2 claim on the Girsanov weights is too strong.
Section 10.1 states that the Girsanov weights, *being* likelihood ratios, "remain bounded in chi-square under refinement",
and the χ^2 row of Table 3 summarises them as bounded. The inference does not hold: being a Radon–Nikodym derivative gives
$\mathbb{E}_P[dQ/dP] = 1$, that is membership in L^1, not in L^2, which is what $\chi^2(Q\|P) = \mathbb{E}_P[(dQ/dP)^2] - 1$ requires.
An elementary counterexample settles it: on $P = \mathrm{Unif}[0,1]$, the density $dQ/dP(x) = \tfrac12 x^{-1/2}$ integrates to 1
and has $\mathbb{E}_P[(dQ/dP)^2] = \tfrac14\int_0^1 x^{-1}dx = \infty$.
For the Girsanov case specifically, the stochastic exponential $\exp(\int\theta\,dW - \tfrac12\int\theta^2dt)$ has second moment
$\mathbb{E}_{Q'}[\exp(\int\theta^2 dt)]$ under the measure with drift 2θ, finite only under a Novikov-type condition on $\int\theta^2 dt$.

The Gaussian control of §9.4 does establish boundednes, for that example. I would restrict the claim to it and name the condition for the general case,
along the lines of: *in the Gaussian control of §9.4 the Girsanov ratio has bounded chi-square divergence under refinement; more generally,*
such boundedness requires a uniform second-moment condition on the Radon–Nikodym densities.
Table 3 should carry the same qualification.

This third item is the only one of the three that does not belong to the present round.

On the backward recursion. Your account is correct and the earlier observation is withdrawn (on verification rather than by concession).
Eliminating the future block by Schur complements gives the precision of the past marginal, whose last entry is the conditional
precision of ξ_m given ξ_{m-1} by the Markov structure; on a chain with $M = 12$ the reciprocal of that entry agrees
with $\operatorname{Var}(\xi_m \mid \xi_{m-1})$ computed from the covariance block to 3.5×10^{-17}.
The recursion does produce forward innovation variances.

In summary, no defect remains in the central construction. Subject to these three amendments, I regard your responses as substantively
satisfactory, and with them implemented the paper will stand in the same position as the previous reports of paper 1,2 and 3.
Item 1 requires your choice of hypothesis; items 2 and 3 are wording.
None affects the conclusion of a proposition, so I do not expect a further round of substantive review.

Ezequiel

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